Final method and research path

Physics core, learned module, and the route from simulation to hardware to the RA-L paper

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(no IMU, real-robot simulation results, shear term, real-time twin, review findings)

1 Decisions in one page

The model is a stack of four layers. Every layer lives at the level of *one module*; the physics assembles any number of modules, any routing and any gravity direction.

Layer	What it is	Learned / identified from	Status
L0 physics core	CAD-based variable-section Euler–Bernoulli module, real 6-channel routing, pull-only tendons as a convex QP, gravity as a vector	nothing (CAD + measured E)	done
L1 module compliance	positive factors on the notched compliances + smooth field (54 numbers)	FEM 1–2 modules (pretraining); 3 global factors re-identified on hardware	done (sim)
L2 hardware physics	tendon stretch (1 number), capstan friction along the path (1–2 numbers), servo calibration	hardware, 1–2 modules	to build
L3 module GP	Gaussian process giving an extra strain field (eigenstrain) per module from its local loads, gravity and motion history, with uncertainty	1–2 modules (FEM first, then hardware)	to build

Input: the tendon displacements $\boldsymbol{\delta\ell}$ sent to the servos, plus the mounting of the base, given once as configuration (it fixes the gravity vector $\boldsymbol{g}$ in the base frame). The twin is sensorless: no IMU or any other sensor on the robot; the base is assumed fixed. Bend angles θ, α appear nowhere in the model or its features.
End point: a real-time digital twin, run live on the lab PC from the servo commands.
Commands: generated exactly as the robot generates them, with the *pay-out mapping* (inside tendons wound in, outside paid out, common mode near minimum compression plus a pretension margin).
Training data: one- and two-module robots only. **Test:** three and four modules, three mountings (upright, horizontal, hanging), other pretension margins and random feasible $\boldsymbol{\delta\ell}$, never seen in training.
Simulation (SOFA FEM) is for method development and pretraining. **Hardware** is for validation; the paper's claims rest on it.

Why this split.

- L0 and L1 generalise by construction: they are properties of the module, and the equilibrium solve handles coupling, slack and gravity for any stack (simulation, real routing, 19 Oct: 0.24 / 0.24 / 0.50 mm on 1 / 2 / 4 modules, training on 1–2; shared routing before: 0.42 / 0.35 / 0.68 mm).
- L2 are effects with known physical form and very few parameters. Learning them as a black box would waste data and break transfer.
- L3 takes only what physics cannot express (nonlinearity at large bends, hysteresis, print variation). It is local to a module and enters the physics as a strain, so composition still holds, and it reports uncertainty.
- Learning on the output of the whole robot is rejected: it does not compose across section counts (simulation: 0.74 mm on 4 modules vs 0.68 for physics alone; whole-robot CC+GP needs 600 four-module poses and reaches 18.5 mm).

2 The robot's real geometry (correction of 10 October)

- Each module has **six channels** every 60° at $r = 9.5$ mm (1.8 mm holes).

- All modules are identical and in the same orientation (notches aligned). The tendons of sections 1–2 use the channels at 0°/120°/240°; the tendons of sections 3–4 use the other three, at 60°/180°/300° (lab frame), running through the free channels of sections 1–2. There is no three-section robot.
- **Consequence:** every 4-section SOFA set so far used the shared routing (all sections on 0/120/240). For the same commands the two routings differ by 115 mm at the tip on average (up to 250 mm). The old 4-section data remain a valid test of the *method* (a robot with another routing), but they cannot validate our robot. All 4-section data must be regenerated with the real routing. 1- and 2-section data are unaffected by the routing; only the three extra holes change the module slightly (bending compliance +1.5–2.4 %, axial +5 % compared with no holes).
- **Already implemented** in `tdcr/eb_prior.py`: per-section tendon azimuths (`routing="paired60"`, or read directly from a cable file), channel holes in the cross-sections, and (for generality) per-module rotation, where the bending compliance becomes a 2×2 tensor. Checked: a rotated module reproduces the rotated shape to 3×10^{-13} mm, and the default reproduces every earlier result exactly.
- **Simulation files** (commit `653e38c`): 6-hole CAD (`cad/std6_tdcr_L150`), mirror-symmetric meshes for 1/2/4 modules, cable files in the lab tendon order with the 0/0/60/60 routing, contact surfaces without the glued joint faces, sweep preset `TDCR_MODEL=std6`, queue `cloud/session/std6_queue.sh`.

Confirmed by Om (10 Oct): 60° routing offset, notches aligned, zero pretension (tensioned by hand until straight and taut), no three-section robot. **Open:** the module joint ("joined by intersection") is modelled as rigid for now.

3 Inputs and commands: tendon displacements and the pay-out mapping

Input. Every model takes the tendon displacements $\boldsymbol{\delta\ell} \in \mathbb{R}^{3N}$ (+ wound in, − paid out), and $\boldsymbol{g}$ when gravity matters. This is what the servos receive, so the twin, the inverse kinematics and the learned parts all work in the robot's own command space. The earlier SOFA designs were drawn in (θ, α) and converted by the constant-curvature mapping; from now on θ, α are not used as inputs anywhere (also not by the baselines, which receive $\boldsymbol{\delta\ell}$ as well).

Pay-out mapping (the robot's actuation). Holding the outside tendons adds compression without adding bend; four sections buckle under the resulting tension. The robot uses the lab's hardware mapping (`tdcr/tendon_mapping.py`): for target bends, $\boldsymbol{\delta\ell} = T(\alpha)\, \boldsymbol{l}_v$ with $l_{v,s} = R_D\, \theta_s$, i.e. pure pay-out (inside wound in, outside paid out by the geometric amounts, no common offset). The robot is tensioned by hand until it stands straight with no slack, so $\boldsymbol{\delta\ell} = \mathbf{0}$ is the taut rest state (zero pretension). In the physics core this leaves the outside tendons slack: the realised bend is smaller than the target, and for intermediate directions it turns towards the taut tendons (one section at 30°, $\alpha = 90°$: realised direction 112°). The models never see the target (θ, α), only $\boldsymbol{\delta\ell}$, so this is learned and predicted, not assumed away.

- **Routing-aware:** a tendon of an upper section also changes length through every module below it, at its own azimuth; the mapping's routing offsets are $\gamma = (0, 0, 60°, 0)$ for the real robot.
- **Generator:** `tools/make_cmd_doe.py` (lab mapping, lab tendon order, physics screening).

Designs (DOE) in command space. Target bends are sampled to cover the workspace and turned into $\boldsymbol{\delta\ell}$ by the pay-out mapping; only $\boldsymbol{\delta\ell}$ is stored and learned. Every design is screened by the physics core before it is simulated: no slack, compression below a buckling margin, bend within the hardware limits. A smaller set of random feasible $\boldsymbol{\delta\ell}$ (inside the same screened region, not produced by the mapping) tests that the model is not tied to one mapping.

4 L0: physics core (final)

Derivation in `docs/eb_prior/eb_prior_tutorial.pdf`; summary of the final form:

1. Cross-sections from the CAD every 0.02 mm, *including the six channel holes*: $A(y)$, $J_x(y)$, $J_z(y)$ in the module frame.
2. Module s rotated by ϕ_s about the axis. Its bending compliance in the base frame is

$$C_s(y) = Q(\phi_s)\, \mathrm{diag}\big(1/J_x(y),\ 1/J_z(y)\big)\, Q(\phi_s)^\top.$$

3. Tendon i at base-frame offset $\boldsymbol{d}_i$ (from the routing), path P_i from the base to the top of its module.
4. Strains: $\varepsilon = \sum_{i \ni y} \lambda_i / A$, $\boldsymbol{\kappa} = C_s(y) \sum_{i \ni y} \lambda_i \boldsymbol{d}_i$ (plus weight and L2/L3 terms).
5. Exact linear tendon map $\Delta\boldsymbol{\ell} = G\boldsymbol{\lambda}$, $G_{ij} = \int_{P_i \cap P_j} \left[1/A + \boldsymbol{d}_i^\top C_s \, \boldsymbol{d}_j\right] dy$, $G \succ 0$.
6. Pull-only tendons: $\min_{\boldsymbol{\lambda} \geq 0} \frac{1}{2} \boldsymbol{\lambda}^\top G \boldsymbol{\lambda} - \boldsymbol{\delta\ell}^\top \boldsymbol{\lambda}$ (NNLS).
7. Gravity $\boldsymbol{g}$ in the base frame, from the known mounting (no sensor): distal weight loads, tendon path shift $\boldsymbol{\delta\ell} \to \boldsymbol{\delta\ell} - \boldsymbol{h}$, fixed-point iteration.
8. Shear (Timoshenko, added 19 Oct): the transverse part of the distal weight shears each cell by $c_S F_\perp / E$, $c_S = 2(1+\nu)/\kappa \int dy/A$ with Cowper's κ for the bored tube. Only gravity across the backbone shears it (tendon loads carry no shear), so G and the QP are unchanged (`tdcr/eb_prior.py`; module-model option `shear="theory"`). Effect: horizontal two modules 1.20 → 1.17 mm; a shear factor tied to the axial notch factor is worse (upright 0.37 → 0.67 mm) and is not used.
9. Shape: exponential-map integration over 0.5 mm cells; positions and frames every 5 mm.

5 L1: module compliance identification (final)

- Parametrisation: $1/A \to e^{s_A c(y) + \theta_A(y)}/A$ and likewise for $1/J_x$, $1/J_z$, with $c(y) = 1$ in notched cells and a smooth field θ (knots every 10 mm, ridge + smoothness penalty). 54 numbers.
- Positive factors $\Rightarrow G \succ 0 \Rightarrow$ the QP keeps a unique solution.
- **Pretraining (FEM):** least squares through the full model on 1- and 2-module FEM poses, with and without gravity. 3-hole meshes: notches 19.7× axial, 5.8 × /5.2× bending compared with plane-section theory at $E = 56$ MPa. **Real robot (std6, 19 Oct, `results/mc_std6.json`, 675 poses, 2.5 min):** 20.2× / 5.9× / 5.3×; held out 0.24 / 0.24 / 0.50 mm on 1 / 2 / 4 modules (gravity-free), upright 0.23 / 0.37, hanging 0.24 / 0.21, horizontal 0.32 / 1.20 ($+x$) and 0.28 / 1.08 mm ($+y$) on 1 / 2 modules. Gate G1′ passed for four modules gravity-free.
- **Horizontal error:** the model reproduces the horizontal two-module sag to 1.2 % (72 mm mean sag), so the error in mm is large only because the sag is. Part of it is a FEM artifact: `UniformMass` puts equal mass on every node, moving weight into the finely meshed notches (centre of mass 80.5 instead of 77.3 mm per module; about 2 % more gravity moment). The identified factors have partly absorbed this, so on hardware the gravity scale must be re-identified. Future SOFA sweeps: `MeshMatrixMass`.
- **Hardware:** keep the FEM shape of $\theta(y)$, re-identify only the three notch factors (s_A, s_X, s_Z) on hardware 1–2 module poses (print and material differ from the FEM).
- **Identifiability rule:** train on ≥ 2 modules (or with external loads); single-module taut data fix only the distribution of the bend, not the stiffness.

6 L2: hardware physics terms (decided after the gap measurement)

- **Tendon stretch:** the servo displacement is the path shortening plus the stretch, $\delta\ell_i = \Delta\ell_i + \lambda_i E \, \ell_i / k_t$, so $G \to G + \mathrm{diag}(E\ell_i/k_t)$. Stays convex. k_t from the tendon datasheet, checked once.
- **Capstan friction along the path:** the tension falls with the wrap angle accumulated from the servo, $T_i(y) = T_i^0 \exp\big(-\sigma_i \, \mu \, \varphi_i(y)\big)$, where $\varphi_i(y)$ is the total turning of the path below y and $\sigma_i = \pm 1$ follows the direction the tendon last moved. This term is the source of hysteresis. It makes the problem non-symmetric, so it is solved by the same fixed-point loop as gravity (friction-free QP, update the load profile, repeat). One parameter μ (Shen et al. 2026 model the same effect with an LCP).
- **Servo calibration:** pulley radius, zero offset and backlash per channel, measured before every campaign.
- Each L2 term is added only if the hardware gap measurement shows it matters (Gate G2).

7 L3: the module-level Gaussian process (final design)

Role. Represent what L0–L2 cannot: nonlinear module behaviour at large bends (in simulation the remaining error doubles above the median bend), hysteresis beyond capstan friction, and print variation.

Output (per module). An *eigenstrain* field $\boldsymbol{\eta}(y) = (\delta\varepsilon, \delta\kappa_{x'}, \delta\kappa_{z'})$ in the module frame. It is added to the strains like a thermal strain. It is parametrised by r modes, $\boldsymbol{\eta}(y) = \sum_{k=1}^{r} a_k \, \boldsymbol{b}_k(y)$, where the $\boldsymbol{b}_k$ come from a PCA of the training residual fields over B-spline coefficients (5 per component; $r \approx 4$–6 expected).

Input (per module, local, the same for any stack).

Feature	Meaning
N_{own}, N_{pass}	axial load from the module's own tendons and from tendons passing through ($/E$)
$M_{x'}$, $M_{z'}$	bending moment carried by the module, in the module frame ($/E$)
$\boldsymbol{g}'$	gravity in the module frame, times module mass ($/E$), 3 numbers
φ_{below}	total bend of the modules below (drives friction), hardware only
σ, ΔM	direction and size of the last load change (hysteresis state), hardware only

All features come from the L0–L2 solution, so they are available for any configuration. The module frame removes the 60° rotation of the upper modules from the inputs. The exact mirror symmetries of the module ($x' \to -x'$, $z' \to -z'$) are used for data augmentation.

Targets.

- *FEM:* the true strains per 5 mm slab follow from consecutive frames, $\boldsymbol{w}_k = \log(R_k^\top R_{k+1})$, and the axial strain from the slab length. The eigenstrain target is the value that makes the model reproduce them. It is found per pose by a small regularised least-squares fit of the mode coefficients through the model ("strain inversion"), so that the tendons' reaction to the eigenstrain is included.
- *Hardware:* the same inversion, but from the measured marker positions along the backbone (markers per module $\geq$ number of modes).

Model. Independent GPs per mode, Matérn-5/2 kernel with automatic relevance determination (one length scale per input), hyperparameters by marginal likelihood. Sparse (inducing-point) GP above about 2000 module samples. One GP shared by all modules; every two-module pose gives two training samples.

First implementation (19 Oct, `analysis/n_transfer.py`). 15 inputs per module (own and passing axial load, moment, the module's bend and end displacement, gravity in its frame and times the modules it carries); one GP with a shared Matérn-5/2 ARD kernel on 12 PCA modes of the per-segment residual (32 segments × 6 = 192 values); 2100 module samples from 1–2 module FEM sets with mirror copies. Two couplings: *output* (added to the physics shape) and *inside* (eigenstrain → tendon path change → QP re-solved). Inside targets come from inverting each residual through the linearised tendon response; rotations must be weighted by a lever arm (100 mm/rad) and the ridge scaled with it, otherwise the targets are wrong. Compared with the section count N and the module position as extra inputs ("local+N") and with no physics inputs ("naive-N"). Smoke tests (old compliance, 300 samples): output coupling helps one module (0.44 → 0.25 mm) but hurts four (1.00 → 1.66 mm); exact inside targets reproduce the training FEM to 0.2 mm, but GP-predicted ones degrade the fit through changes of the slack set. Full run on the real-robot data (19 Oct, `results/n_transfer.json`, 2100 samples, 12 modes): physics + L1 alone is best on four modules (0.50 mm); every learned correction is equal or worse there (output: naive-N 0.54, local+N 1.02, local 1.21 mm; inside: 2.6–5.9 mm). Output coupling helps only one module (0.24 → 0.13 mm, 0.37° → 0.10°); N as an input brings nothing (its length scale is irrelevant). Exact inside targets lower the training error only from 0.24 to 0.22 mm: most of the residual lies in directions the held tendons cancel. The residual after L1 (∼0.24 mm) is below the FEM's mesh error (∼0.6 mm), so in simulation L3 fits discretisation noise. **Decision:** the simulation claims rest on L0+L1; L3 is justified only by hardware effects (Gate G2) and, if used, is gated by its own uncertainty so that it falls back to the physics.

Inference (inside the physics).

1. Solve L0–L2 ⇒ tensions and loads ⇒ features per module.
2. GP ⇒ mode coefficients ⇒ eigenstrain $\boldsymbol{\eta}(y)$ in every module.
3. The eigenstrain changes the tendon paths by $h_i = \int_{P_i} (\delta\varepsilon + \delta\boldsymbol{\kappa} \cdot \boldsymbol{d}_i)\, dy$: re-solve the QP with $\boldsymbol{\delta\ell} - \boldsymbol{h}$ (as for gravity).
4. Repeat until the loads stop changing (expected 2–4 rounds). Integrate the shape.

Uncertainty. Draw 32 samples of the mode coefficients from the GP posterior and push them through steps 3–4 ⇒ covariance of every station position and orientation. Report the coverage of the 95 % bands.

Why a GP. The data are small (hundreds of module samples on hardware), the inputs low-dimensional, and the paper needs calibrated uncertainty. In simulation a GP and a neural-network ensemble were equally accurate (2.79 vs 2.57 mm, 0.17 vs 0.16 mm), so the learner is not the bottleneck. If the data grow, the same slot can take a learned convex module energy (input-convex network) without changing the rest.

8 Evaluation protocol

Generalisation splits (train on the left column only):

Axis	Train	Test
Section count	1 and 2 modules	4 modules (real routing)
Mounting	upright	horizontal (two directions), hanging
Actuation	pay-out mapping, nominal margin	other pretension margins; random feasible $\boldsymbol{\delta\ell}$
Sim → real	FEM (pretraining) + 1–2 hardware modules	hardware 1–4 modules, all mountings

Metrics: tip position (mm, % of length), tip orientation (geodesic angle), mean error along the backbone, 95th percentiles, runtime, training poses needed, coverage of the uncertainty bands; for mountings, the error also relative to the gravity deflection; for the twin, update rate and latency.

Baselines:

- constant curvature;
- CC + GP on the whole robot (trained per configuration, the usual practice);
- the mechanics prior (Camarillo-type) and mechanics + GP;
- a pure GP or network from commands;
- ablations of ours (L0 only; L0+L1; L0+L1+L2; with L3 on the output instead of inside; one-module training only);
- a Cosserat model with prescribed displacements and slack (Oliver-Butler et al. 2019): the accuracy reference and the runtime contrast for the real-time claim (real-time Cosserat dynamics: Till, Aloi & Rucker 2019);
- a learned compositional baseline (one shared per-module model chained along the stack) and the module GP with N as an input.

9 The path

Phase A now, on this machine (no SOFA)

A1 Prior generalised to the real routing and holes. **done**

A2 Strain-inversion tool: eigenstrain targets from FEM frames (and later from markers).

A3 Module GP (L3) on the existing shared-routing data: train on 1–2 modules, test on the old 4-module set (a valid robot with another routing).

A4 Command generator in $\boldsymbol{\delta\ell}$ space: the robot's pay-out mapping, routing-aware (paired routing), common mode from the physics core, with physics screening (slack, compression, bend limits); plus the random-feasible-$\boldsymbol{\delta\ell}$ generator. Replaces the (θ, α) designs.

Gate G1. L3 must improve held-out 4 modules beyond L0+L1 (0.68 mm) and give calibrated bands. If not, L3 stays only for hardware effects and the simulation story uses L0+L1.

Phase B lab PC: regenerate the simulations with the real robot

B1 CAD of the 6-hole module (`cad/std6_tdcr_L150`). **done**

B2 Mirror-symmetric tetrahedral mesh (gmsh, as before), stacked for 2 and 4 modules (identical modules, no rotation). **done**

B3 Cable files with the 0/0/60/60 routing in the lab tendon order; contact surfaces without joint faces. **done**

B4 Sweeps (the existing queue scripts), all with $E = 56$ MPa, $\nu = 0.48$, every command from the pay-out mapping (inputs stored as $\boldsymbol{\delta\ell}$):

set	1 module	2 modules	4 modules
train, gravity-free	150	200	–
train, upright	150	200	–
test, gravity-free	60	100	100
test, upright	60	100	100
test, horizontal ($+x$, $+y$)	60	100	100
test, hanging	60	100	100
test, random feasible $\boldsymbol{\delta\ell}$	60	100	100

B5 Re-identify L1 on the new 1–2 module sets; evaluate all splits; retrain L3; regenerate Figs. 4–6 and Tables I–II of the outline.

Status 19 Oct. B4 ran on 11 Oct except the 4-module mounting sets and the 2/4-module random sets (the 2-module horizontal $+y$ test has 22 of 100 poses). B5: L1 re-identified (`results/mc_std6.json`); evaluation code fixed first (the loader gave the real-robot sets the shared routing, no holes and upright gravity for every mounting: 257 mm instead of 0.90 mm on four modules; the mirror augmentation swapped the wrong tendons in sections 3–4). L3 comparison running.

Gate G1′. The full model trained on 1–2 modules must stay sub-millimetre on 4 modules and on all mountings in FEM before hardware time is spent. *19 Oct:* passed for four modules gravity-free (0.50 mm, 95th percentile 1.01 mm) and for 1–2 modules upright and hanging; horizontal two modules 1.08–1.20 mm (1.2 % of the sag, partly the FEM mass artifact); four-module mountings not yet simulated.

Phase C hardware

C0 Before printing the full set: print four modules and check that the stack stands upright under its own weight (self-weight estimate with slack tendons: about 5× over the buckling limit for four modules, 0.65× for two); if not, change the module design now or limit four-module tests to hanging. Print at least 5–6 modules from one batch and 2 from another (tests on unseen prints). Bench tests per module: axial and bending stiffness (hanging weights), mass.

C1 Setup: rigid mounts at fixed angles (0°, 90°, 180°; no IMU: the mounting is configuration, checked once at setup with the motion-capture base markers, to about 1°), servo calibration, tendon homing to a defined small tension (servo current or inline load cell; a 0.2 mm zero error tilts a module by 0.8°, about 8 mm at a four-module tip), ground truth by camera/markers (at least 4 markers per module, 0.2 mm accuracy target), optional tendon load cells.

C2 **Gap measurement:** 1–2 modules, about 50 poses, compare L0+L1 (FEM-pretrained) with the measured shape.

C3 Identification on 1–2 modules: L1 global factors, L2 terms as needed, L3 GP; include repeated loading/unloading cycles for hysteresis.

C4 Tests: 4 modules × upright / horizontal / hanging, pay-out commands and random feasible $\boldsymbol{\delta\ell}$, never trained on.

C5 Uncertainty calibration and runtime on the lab PC.

C6 The twin live: servo positions streamed into the twin; update rate and latency for 1 / 2 / 4 modules; slow continuous trajectories against motion capture; one live use (buckling warning or reaching a target).

Gate G2 (after C2). Gap < 2 mm: the paper leads with physics transfer, and L3 handles hysteresis. Gap ≥ 2 mm: L2 and L3 carry the result, with the ablations showing how much each layer contributes.

Phase D paper

Figures and tables of the outline (`docs/paper_outline`), writing, internal review with Prof. Vadali.

10 Risks and mitigations

Risk	Mitigation
Four-module FEM poses buckle or snap	pay-out mapping (about half the compression); physics screening of every design; report the stable workspace
Friction dominates on hardware	L2 capstan term, history features in L3, lower tensions from the pay-out mapping
Marker occlusion or too few markers for the inversion	fewer modes; regularised inversion; tip pose as a minimum target
Hardware lead time	Phases A and B run in parallel with C1; the simulation results already form most of Tables I–II
Novelty challenged (physics known)	claim the configuration-independent learning and hardware generalisation, not the beam theory; cite Camarillo 2009, Oliver-Butler 2019, Ménager 2023, Gao 2024, Sue et al. 2026 (modular, sensorised)
Four-module stack collapses under its own weight	check with four printed modules before the campaign; redesign or limit to hanging
Tendon zero offsets dominate the hardware error	tension-based homing; identify offsets as L2 parameters
FEM artefacts identified as physics (per-node mass, mesh error $\sim$0.6 mm)	re-identify the gravity scale on hardware; `MeshMatrixMass` in new sweeps
Test sets screened by the model under test	add unscreened or FEM-screened tests; report workspace coverage
Mounting angle wrong	fixed-angle mounts, checked once at setup; report the sensitivity
Real-time claim challenged	measure rate and latency live, with L3 and gravity; compare with a Cosserat reference